

Luis Arturo González Salas

M.Sc. in Robotics Engineering

Valid residence permit to work in France. Open to international mobility.

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Education

JEMARO Double Degree (Japan European Master in Advance Robotics)

Keio University – M.Sc. in Science & Engineering (Tokyo, Japan)

2024-2025

Ecole Centrale de Nantes – M.Sc. in Advanced Robotics (Nantes, France)

2023-2024

Tecnológico de Monterrey (Monterrey, Mexico)

B.Sc. in Mechatronics Engineering – Minor in Cyber-physical Systems

2019-2023

Academic Projects

- **Thesis “Micro-scale Particle Manipulation Using Bilateral Control and Optical Tweezers: A 2-Axis Approach”:** Designed a two-axis haptic teleoperation system for tactile control of optically trapped microparticles. Integrated experimental trap stiffness calibration with admittance control and reaction force observers using C++ and Python.
- **Simulated a Hand Guided Cobotic Assistant for Percutaneous Interventions:** Simulation of hand guidance of a redundant macro-mini system with self-positioning of the macro’s end-effector based on the micro’s end-effector.
- **Automated Production Line:** Automated a keychain production plant by developing and integrating a mobile collaborative robot system with digital twin technology. Coordinated the navigation, task execution, and safety protocols of the mobile robots and collaborative arms using ROS and industrial controllers.
- **Robotic Welding Cell:** Designed an automated robotic work cell for a laser welding production line. Integrated ABB industrial robots, custom fixtures, and safety systems within RobotStudio for offline simulation and collision detection
- **Automatic Tire Inflation System:** Created an RTOS driven embedded system prototype for John Deere to regulate tractor tire pressure dynamically. Featured PID control and active leakage compensation to enhance vehicle adaptability.

Code Repositories

2026

- **ROS 2 Sorting Architecture:** Built a ROS 2 and Gazebo simulation of a robotic arm environment. The system used stereo vision to detect targets and an inverse kinematics solver for autonomous pick and place sorting using Python and C++.
- **Autonomous Swerve Mobile Robot:** Created a ROS 2 stack for a 4 wheel swerve robot using Python and C++. Implemented custom kinematics and RRT algorithm to map unknown environments autonomously in Gazebo using Nav2 and SLAM.

Work Experience

Engineering Project Management Intern at Navistar International (Monterrey, Mexico)

2022-2023

- Programmed a platform to control, review and summarize company projects using Visual Basic.
- Participated in daily standups and design reviews with coworkers from different cities in the USA.

Freelance Web Developer (Rotary Club– Retail Store)

2019-2021

- Developed a full-stack web application backed by a custom relational database, featuring user authentication, role-based access, and a dynamic analytics dashboard to manage inventory, sales, and expenditures.
- Built a responsive visitor interface and interactive digital catalog equipped with a real-time search engine to enhance product discoverability and properly showcase projects, calendars, and partners.

Main Achievements

- Received Erasmus Mundus Scholarship for master studies. 2023
- Achieved CSWA Solidworks certification. 2021
- Received full coverage scholarship tuition for bachelor studies. 2019
- Won Silver (National) and 2 Gold (Veracruz) medals at the Olympiads of Informatics. 2016 - 2018

Skills

Programming Languages: C++, Python, MATLAB, Structured Text, MySQL

Software & Tools: Linux/Ubuntu, Git, ROS / ROS 2 (ros2_control, tf2, MoveIt 2, RViz), Gazebo, OpenCV, TensorFlow, PyTorch, TIA Portal, Siemens NX, RobotStudio, Tecnomatix, SolidWorks, MATLAB, Simulink, STM32 (Cube IDE, RTOS), Visual Basic, WandB, Gantt Project, Office Package, CI/CD (GitHub Actions)

Languages: Spanish (Native), English (Fluent), French (Intermediate), German (Intermediate)

Soft skills: High Learning Velocity, Hands-on Mentality, Interdisciplinary Team Collaboration, Creative Problem-Solving, Critical Thinking, Adaptability, Resilience